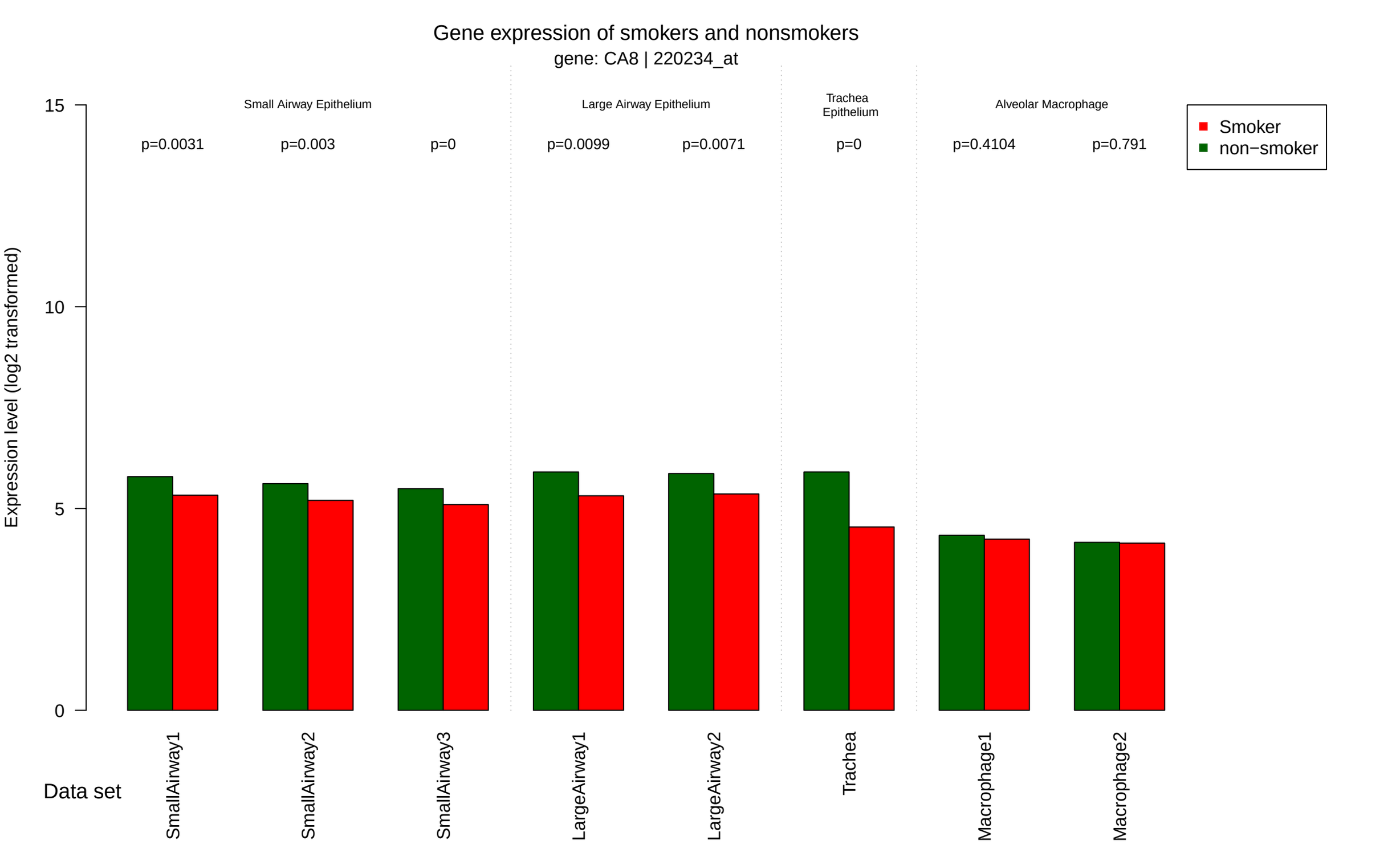
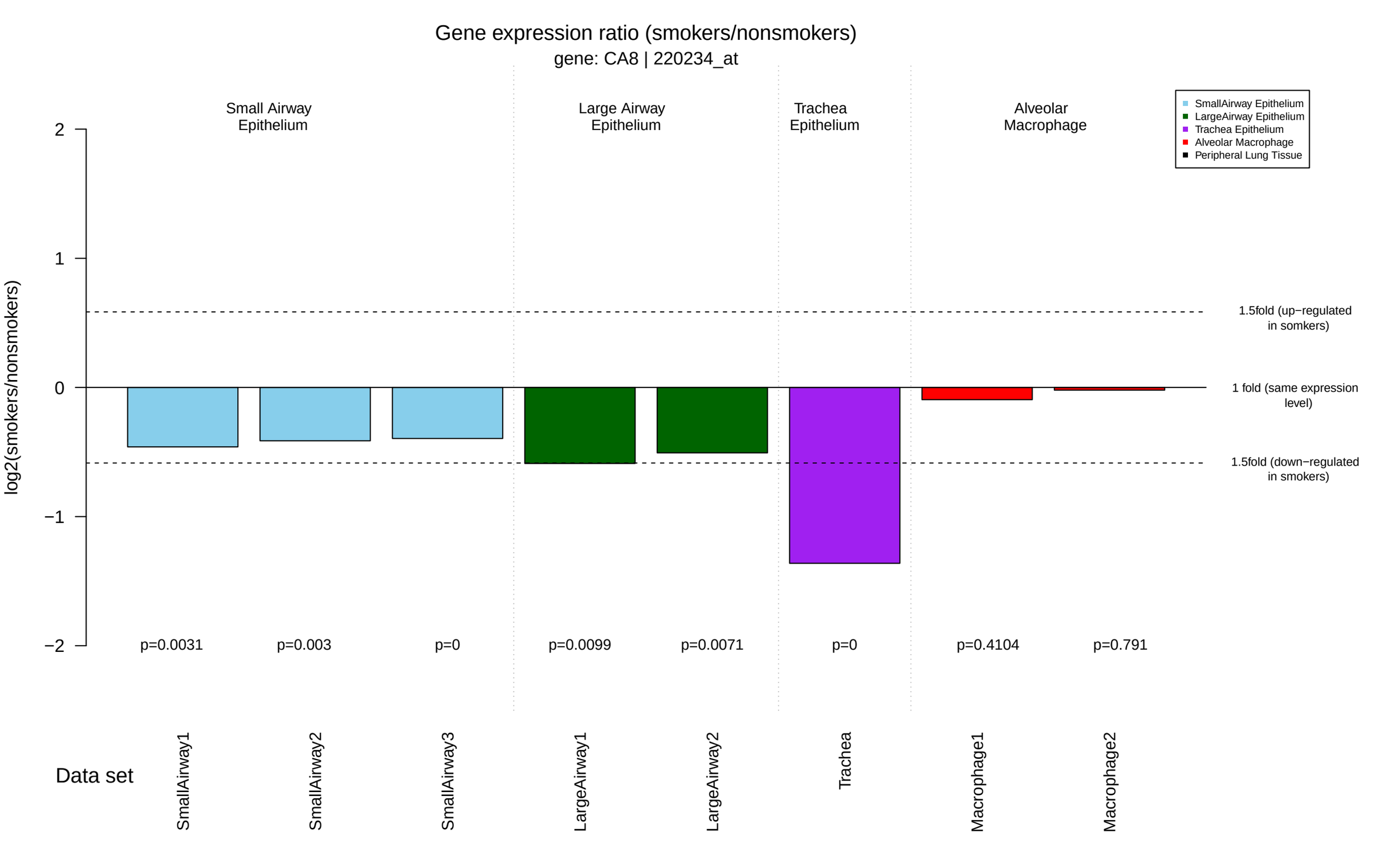
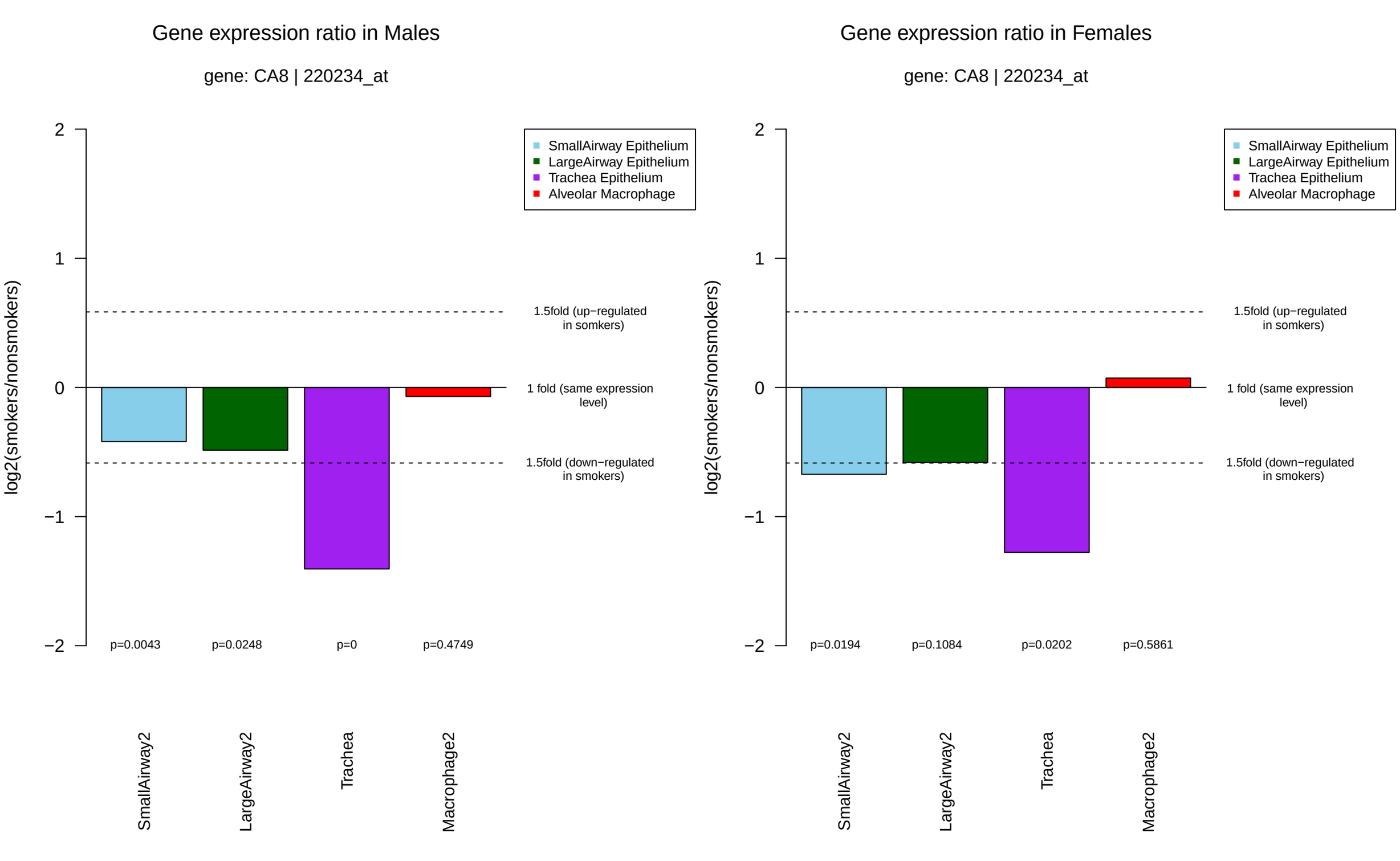
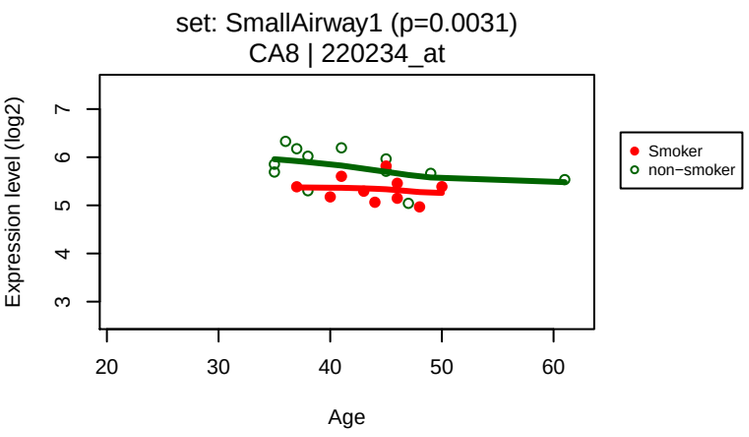
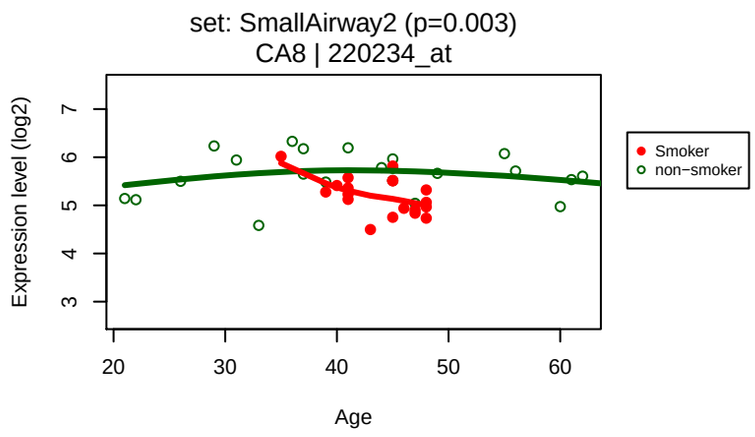
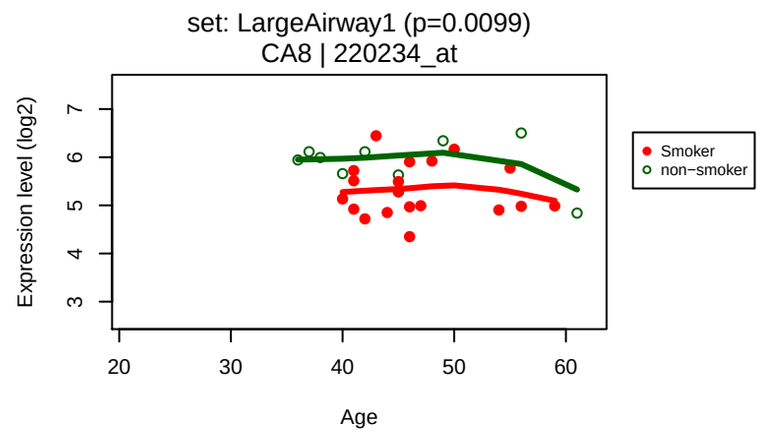

Gene expression of smokers and nonsmokers
gene: CA8 | 220234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0031
p=0.003
p=0
p=0.0099
p=0.0071
p=0
p=0.4104
p=0.791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

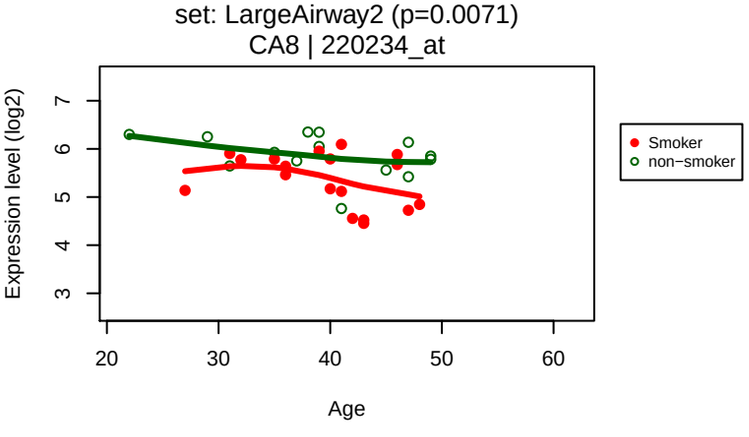

Gene expression ratio (smokers/nonsmokers)
gene: CA8 | 220234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.003
p=0
p=0.0099
p=0.0071
p=0
p=0.4104
p=0.791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

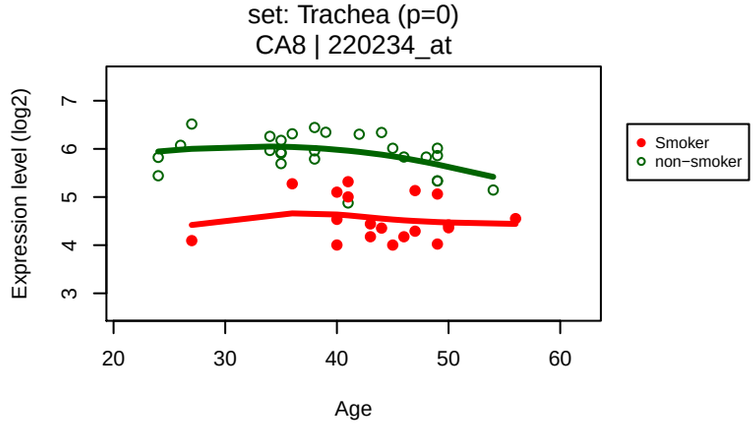

Gene expression ratio in Males
gene: CA8 | 220234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0043
p=0.0248
p=0
p=0.4749
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA8 | 220234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0194
p=0.1084
p=0.0202
p=0.5861
SmallAirway2
LargeAirway2
Trachea
Macrophage2

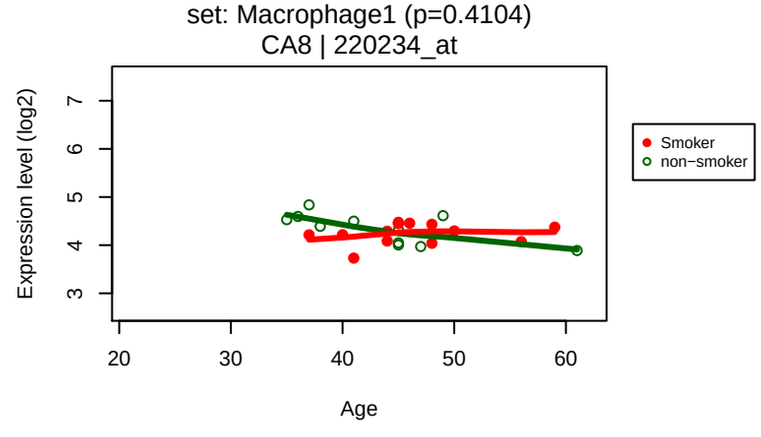

set: SmallAirway1 (p=0.0031)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker

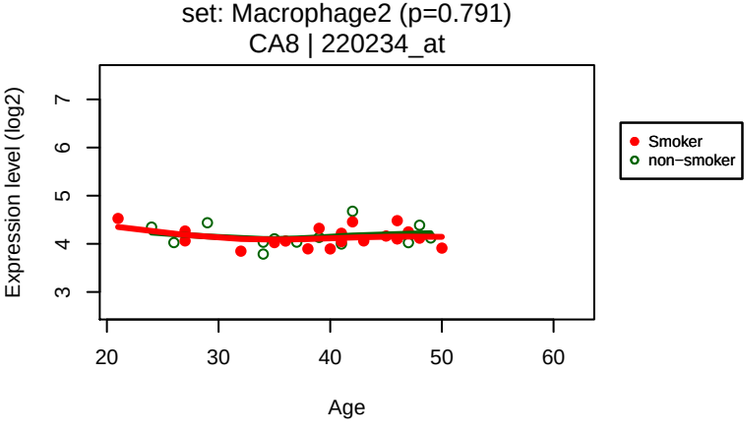

set: SmallAirway2 (p=0.003)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker

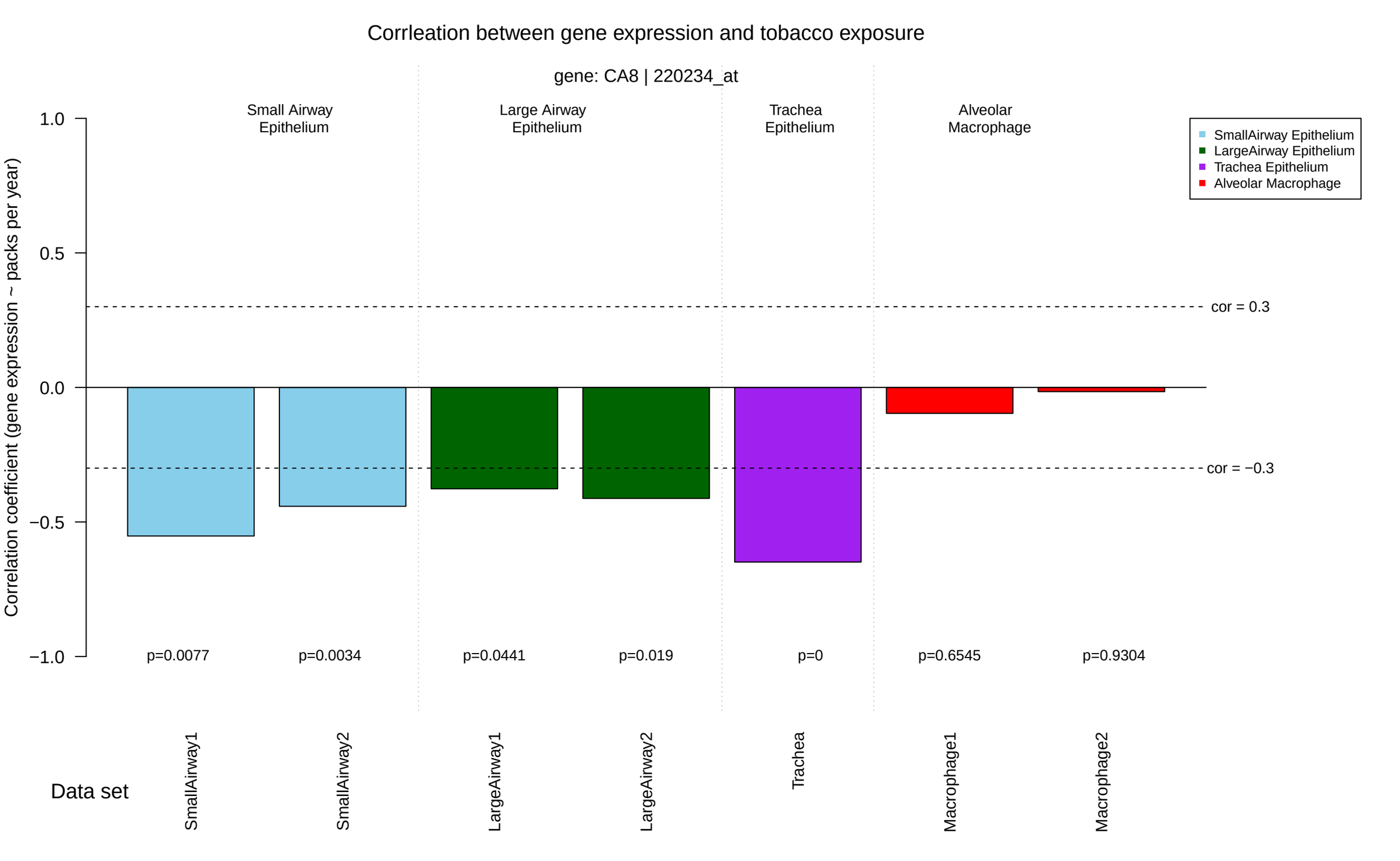

set: LargeAirway1 (p=0.0099)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0071)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4104)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.791)
CA8 | 220234_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA8 | 220234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0077
p=0.0034
p=0.0441
p=0.019
p=0
p=0.6545
p=0.9304
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2